L-F-1593
July 7, 1970

SUPERSEDING
Fed. Spec. L-F-750A
March 1, 1967
MIL-F-43523
1 August 1967

FEDERAL SPECIFICATION

FUNNELS, PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic funnels.

1.2 Classification.

1.2.1 Classes and sizes. Funnel shall be of the following classes and sizes (capacities) as specified (see 6.2).

CLASS 1 - Polypropylene

CAPACITY - 1 pint, 5 to 5-3/8 inch OD
CAPACITY - 4 quart, 10 to 12 inch OD

CLASS 2 - Polyethylene

CAPACITY - 4 ounce, 2-7/8 to 3-7/16 inch OD
CAPACITY - 1/2 pint, 3-7/8 to 4-1/4 inch OD
CAPACITY - 1 pint, 5 to 5-3/8 inch OD
CAPACITY - 1 quart, 6-3/8 to 7-3/4 inch OD
CAPACITY - 2 quart, 7 to 9 inch OD
CAPACITY - 4 quart, 10 to 12 inch OD

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 7240

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Federal Specifications:

L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-676	-	Boxes, Setup.
PPP-B-636	-	Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

ederal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-22748	-	Plastic, Polyethylene and Copolymers, High Density, Molding and Extrusion Material.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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U.S. Department of Health, Education and Welfare:

:de... Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

.pplication for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

National Classification Board:

National Motor Freight Classification.

.pplication for copies should be addressed to American Trucking Associations, :: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

pplication for copies should be addressed to the Uniform Classification ittee, 202 Union Station, Chicago, Illinois 60606.)

REQUIREMENTS

1 Material. The materials used shall conform to the applicable specifica- s requirements specified, unless otherwise indicated herein. Material de...cely specified shall be of the quality and type normally used for the ose in commercial practice.

1.1 Plastic. The items shall be made from virgin plastic materials as ified. Clean, unburned plastic material in the form of imperfect parts, es, runners, or other scrap of the same composition as the virgin material, produced in the molding or finishing operation, may be reground and mixed the material specified. Items made by rotational molding shall be made virgin material only. Materials used in the manufacture of funnels shall orm to the Federal Food, Drug and Cosmetic Act, Food Additive Amendment, ode Federal Regulation 121.2510 for contact with food.

1.1.1 High density polyethylene. High density polyethylene for Class 2 els to class A, grade 1, 3 or 5 of MIL-P-22748.

..1.2 Polypropylene. Polypropylene for Class 1 funnels shall conform to I of L-P-394.

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3.2 Construction. The funnels covered by this specification shall be of the ain type with neither a shallow nor deep rim, without a strainer, and shall meet the requirements specified herein. The funnels shall be of one piece, seamless construction with a conical body and a straight sided tapered spout. The top edge shall be rounded and stiffened.

3.3 Color. The funnel shall be the natural color of the polyethylene or polypropylene used.

3.4 Surfaces. The surfaces of the funnel shall be uniform and free from blisters, mold marks, weld marks, cracks, or lumps. All flash shall be completely removed from surfaces, openings and holes.

3.5 Dimensions, composition, and capacities. The funnels shall conform to the requirements specified in table I and have an inside angle of $60\frac{1}{4}^{\circ} \pm 5\frac{1}{4}^{\circ}$ in the body portion. The nominal capacity of the funnels shall be as specified when tested in accordance with 4.3. Funnels up to 1 pint nominal capacity may have a minus 10 percent tolerance and funnels 1 quart and over capacity may have a minus 3 percent tolerance.

TABLE I. Capacity and dimensions

Nominal capacity (sizes)	Dimensions (inches)			
	Top overall diameter	Tip overall diameter	Overall height	Thickness (min)
oz.	2-7/8 to 3-1/4	5/16 ± 1/16	3 ± 1/4	0.050
1/2 pint	3-7/8 to 4-1/4	3/8 ± 1/16	4 ± 1/4	0.055
1 pint	5 to 5-1/4	1/2 ± 1/8	5 ± 1/4	0.060
1 quart	6-3/8 to 7-3/4	5/8 ± 1/8	7 ± 3/4	0.070
2 quart	7 to 9	3/4 ± 1/8	8 ± 1	0.090
4 quart	10 to 12	1 ± 3/16	11 ± 1-1/4	0.120

3.6 Workmanship. The funnels shall conform to the quality established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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Certificate of compliance. Where certificates of compliance are sub-t..., the Government reserves the right to check test such items to determine e validity of the certification.

4.2 Inspection. Sampling for inspection and acceptance shall be in accord-ce with the provisions set forth in MIL-STD-105, except where otherwise dicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, mponents and materials shall be tested in accordance with all the requirements referenced specifications, drawings and standards unless otherwise excluded, ended or qualified in this specification or applicable purchase documents. ems listed below may be accepted on the basis of a supplier's certificate compliance.

Items	Requirement paragraph
Conformance of materials to Food, Drug and Cosmetic Act	3.1.1
Plastic material	3.1.1.1 or 3.1.1.2 (as applicable)

.2.2 Inspection of the end item.

1 Visual examination. Examination of the end item shall be in accord-e...n the defects listed below and the acceptable quality levels (AQLs) set th in 4.2.2.4. For purposes of sampling, all funnels of the same class and acity offered for delivery at one time, shall be considered a lot. The ple unit shall be one funnel.

mine	Defect
ish	Not uniform, contains blister, mold mark, weld mark, crack or lump. Contains flash.
struction and rkmanship	Not design specified. Top edges not rounded and strengthened. Not one piece construction. Body not conical. Sides not straight and tapered on spout.

.2.2.2 Dimensional examination. Inspection shall be made of funnels to ermine compliance with dimensions of funnels as indicated in table I. Any ension not within specified tolerances shall be classified as a defect.

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4.2.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine that marking, packaging, packing, and contents comply with the requirements of section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be sealed. Defects of closure listed below shall be examined on containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.2.2.4 Inspection levels and acceptable quality levels (AQLs). The inspection levels, for purposes of determining the sample size, and the AQLs expressed defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1	II	6.5
4.2.2.2	S-2	4.0
4.2.2.3	S-2	2.5

4.3 Testing of the end item. The end item shall be tested for capacity as specified in 3.5 and 4.4. The sample unit shall be one funnel. The inspection level shall be S.2 with an AQL of 4.0 defects per 100 units.

4.4 Test method.

4.4.1 Capacity test. The funnels shall be tested for minimum capacity by covering the end of the spout and pouring the minimum capacity of water into the funnel as specified in 3.5. The inability of the funnel to contain the minimum capacity shall constitute a failure.

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PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Funnels, of one class and capacity only, shall be nested and packaged inverted in the quantities as specified in table II, within a snug-fitting fiberboard box conforming to type CF, variety SW, class domestic, grade 200, style RSC of PPP-B-636. Alternatively, 4-ounce capacity funnels shall be packaged in a folding or setup paperboard box conforming to PPP-B-566 or PPP-B-676, or in sealed polyethylene bag. Each box shall be closed in accordance with the appendix of the applicable box specification.

TABLE II. Packaging and packing quantity

Funnel capacity	Quantity per package	Quantity per shipping container
4 ounce	12	144
1/2 pint	12	144
1 pint	12	144
1 quart	12	72
2 quart	--	12
1 gallon	--	12

5.1.2 Level C. Funnels shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Funnels of one class and capacity only, packaged as specified in 5.1, shall be packed in the quantities as specified in table II in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Containers for funnels of 2-quart, 1-gallon, or polyethylene bagged 4-ounce capacity shall be provided with a fitted, full-height taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Funnels of one class and capacity only, packaged as specified in 5.1, shall be packed in the quantities as specified in table II in a snug-fitting fiberboard shipping container conforming to type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Contianers for funnels of 2-quart,

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lon, or polyethylene bagged 4-ounce capacity shall be provided with a fitted, full-height taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Funnels, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or Fed. Std. No. 123, as applicable.

6. NOTES

6.1 Intended use. The class 1 (polypropylene) funnels are intended for use in areas where they may be exposed to instances of relatively high temperatures, such as in kitchens and hospitals. The class 2 funnels are intended for general u

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and capacity of funnels desired (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 This specification includes the requirements of MIL-F-43523, dated 1 August 1967, and L-F-750A, dated March 1, 1967.

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ARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD
Navy - YD
Air Force - 84

User activities:

Navy - MS, MC

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

GSA

Project No. 7240-0137

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SPECIFICATION COMMENT SHEET

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountere in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, **and mail to: General Services Administration, FSS (FMSH), Wash., D. C. 2040**

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the docum or serve to amend contractual requirements.

1. SPECIFICATION

L-F-1593 Funnels, plastic

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in ta do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

FORM GSA FEB 69 2:

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